Integrated Design and Fabrication Representations

Lecture 1.0 | 19 October 2026

Learning Goals

1. Understand the role of **geometric representation** as the common core of computational design.
2. Gain familiarity with the concepts of **parametric, algorithmic** and **computational design**, and how they build on each other.
3. Understand how **materials and fabrication processes** inform the choice of **integrated design representations**.

Contents

1 Motivation

2 Representation as the core of Parametric, Algorithmic and Computational Design

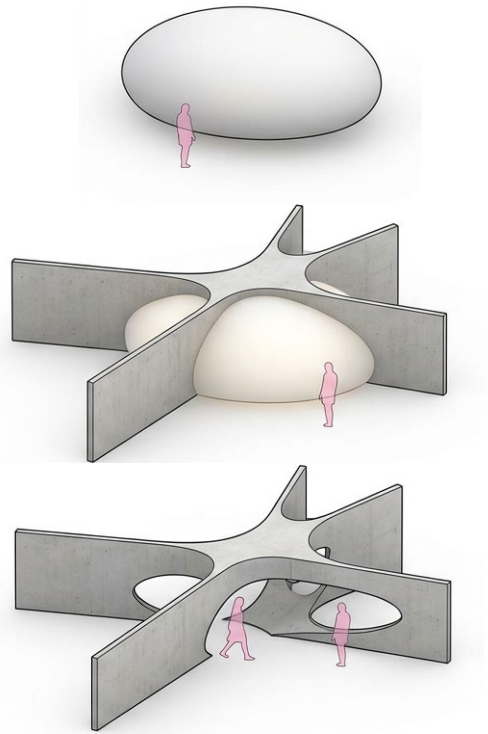
3 Integrated Design and Fabrication Representations

4 Outlook

Digital tool-driven Design



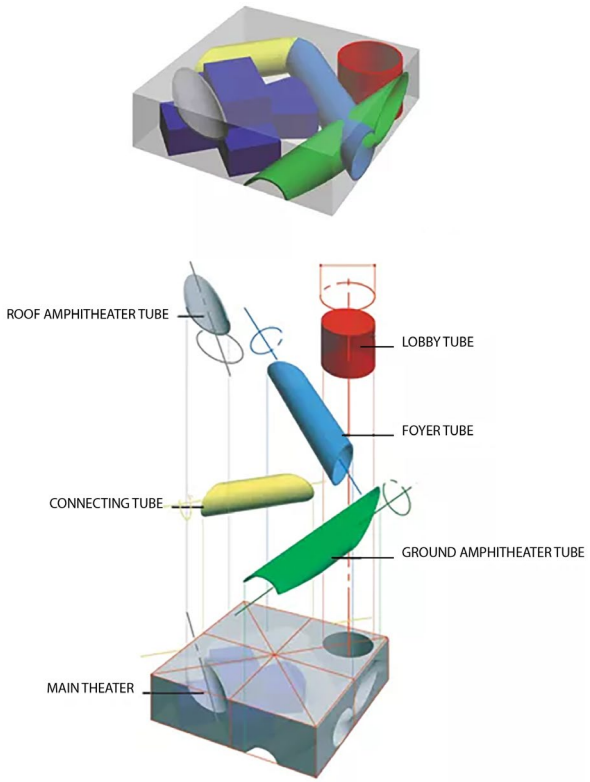
Source: Höweler and Yoon Architecture, LLP



BOOLEAN DIFFERENCE

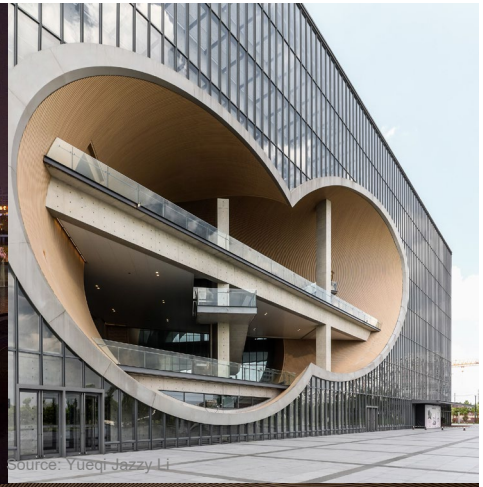
Sean Collier Memorial at MIT, 2015

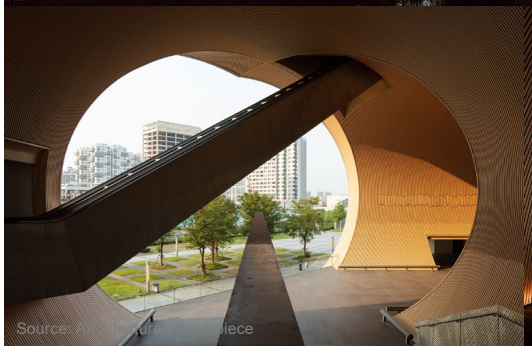
Digital tool-driven Design

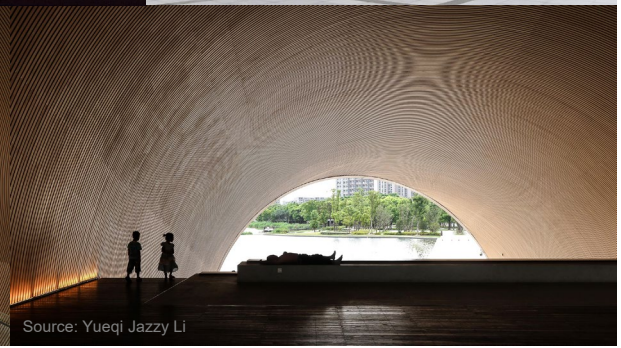


Poly Grand, Shanghai, Tadao Ando, 2014





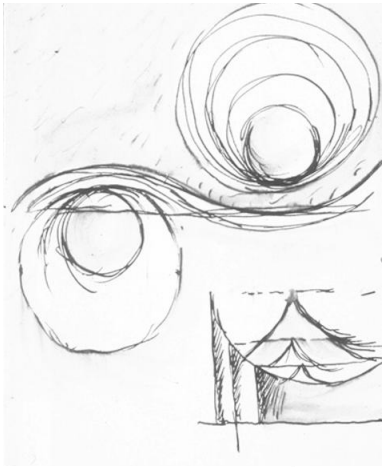




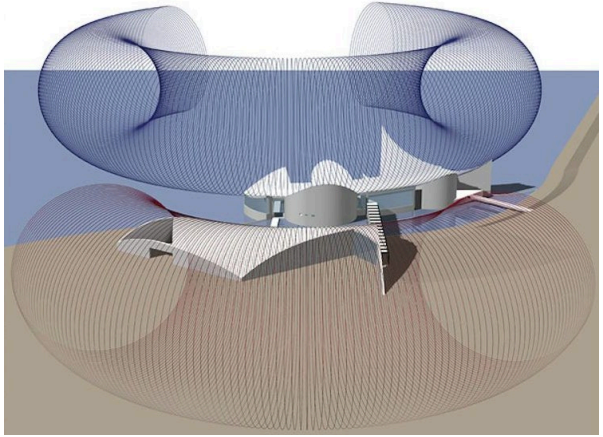
Digital tool-driven Design













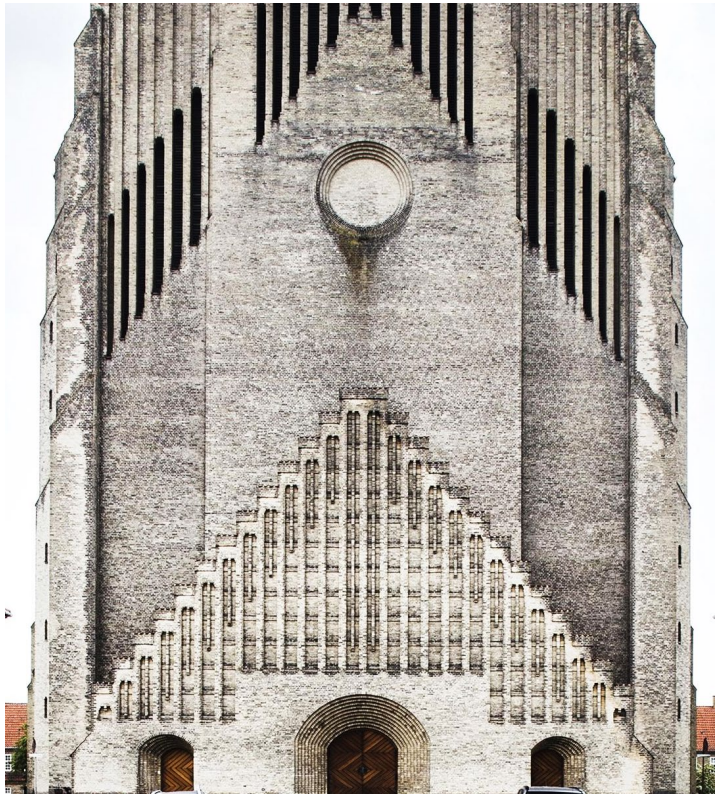
Exploration Place, Moshe Safdie, 2000

Material-aware Design

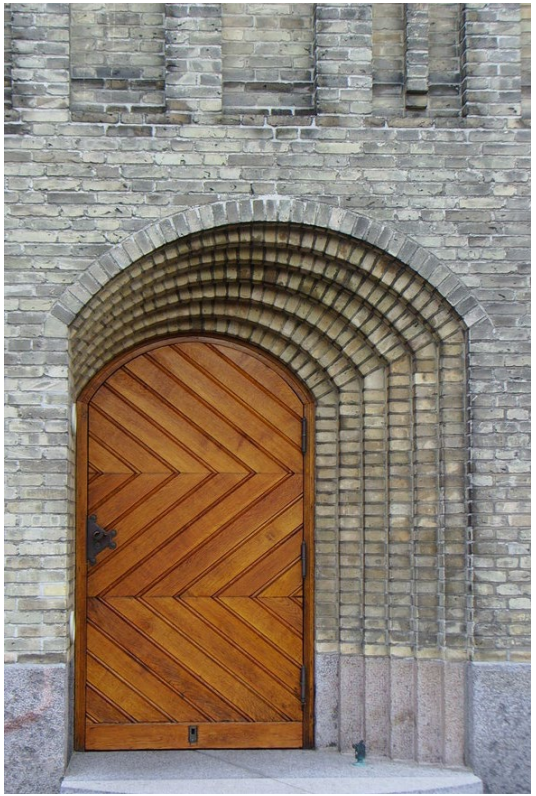


Grundtvig’s Church in Copenhagen, Denmark, Peder Vilhelm Jensen-Klint, 1940

Material-aware Design







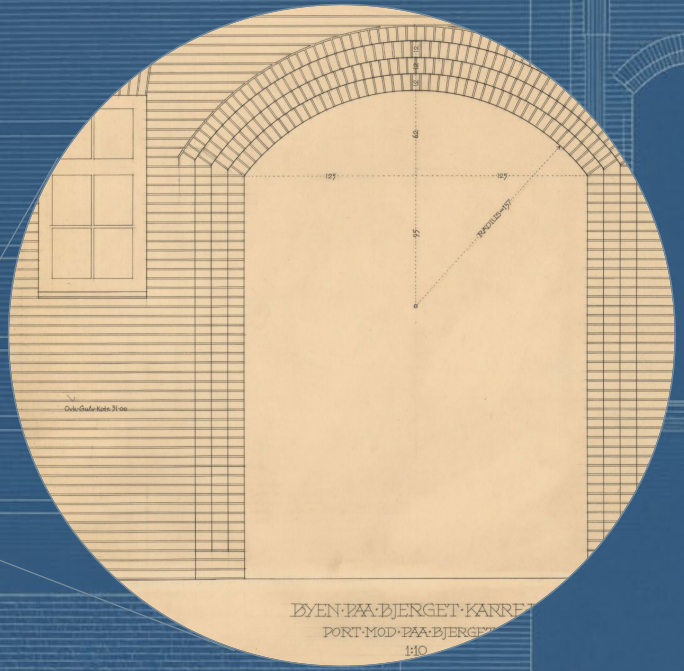
Grundtvig's Church in Copenhagen, Denmark, Peder Vilhelm Jensen-Klint, 1940

Geometry as Spatial Description



Geometry as a **representational tool** for spatial planning, proportion, and design intent.

Geometry for Procedural Description



Geometry as a **prescriptive tool** for fabrication, detailing, tolerances and coordination.

Geometric representation is the core

DIGITAL DESIGN

Operations on *Form*

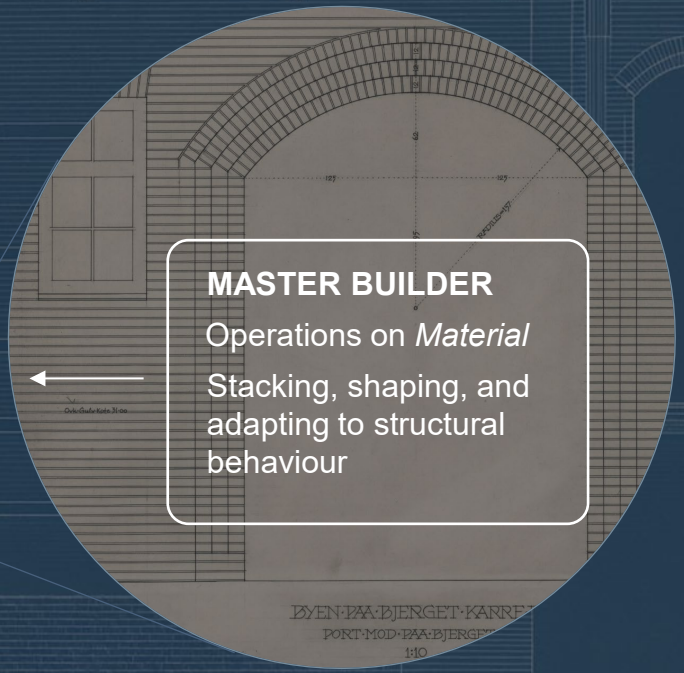
Subtracting, intersecting, and combining volumes

Geometric representation

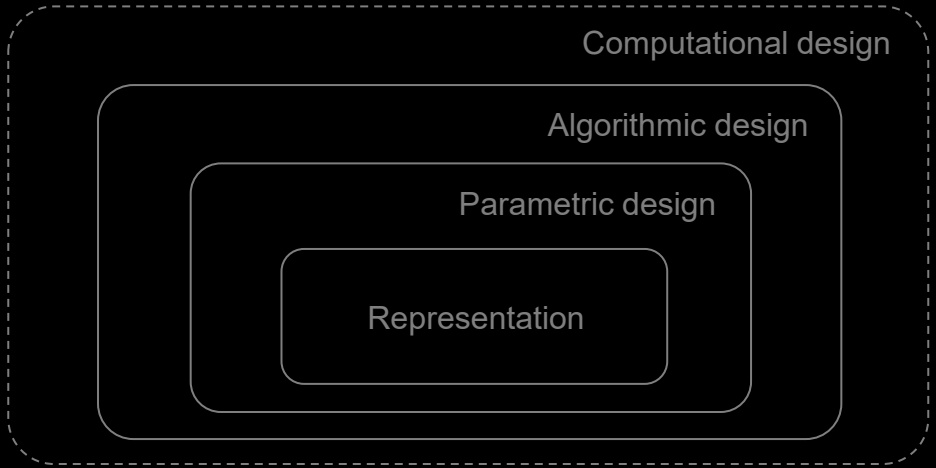
MASTER BUILDER

Operations on *Material*

Stacking, shaping, and adapting to structural behaviour



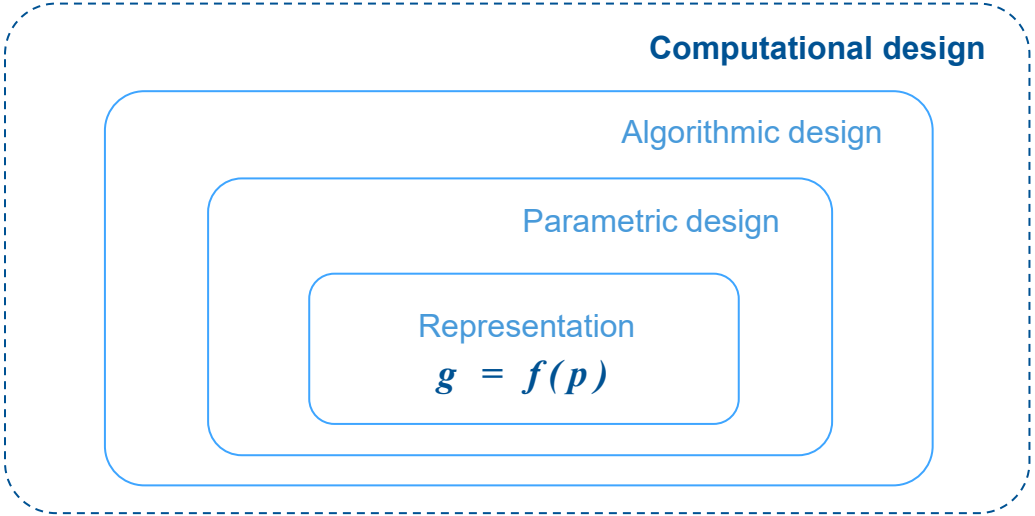
How can geometry and construction be integrated in a meaningful way?



2 /

Representation as the core of Parametric, Algorithmic and Computational Design

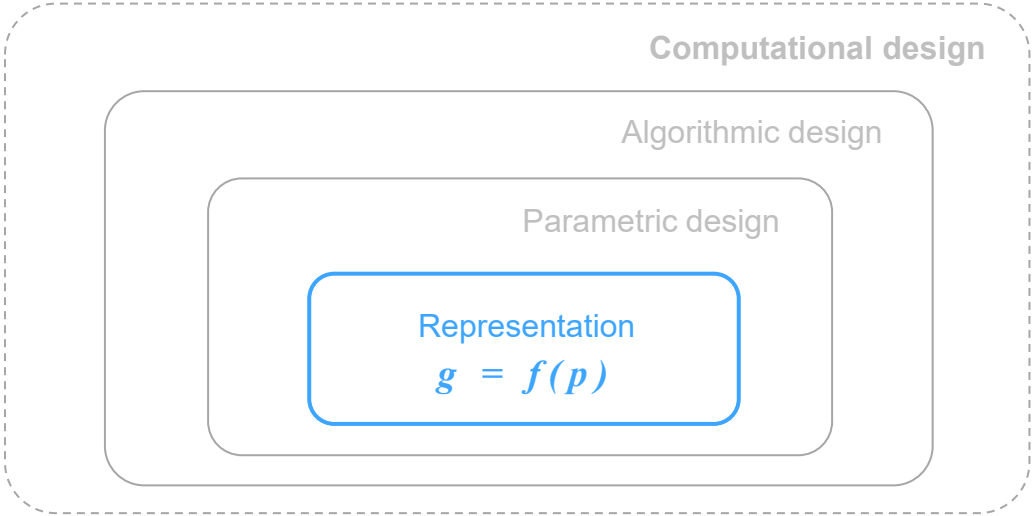
Computational design



Design approach that uses **computation** to generate, transform, evaluate, and refine design solutions through formalized relationships, quantifiable properties, and algorithmic processes.

Source: Terzidis, K.: Algorithmic Architecture, 2006.

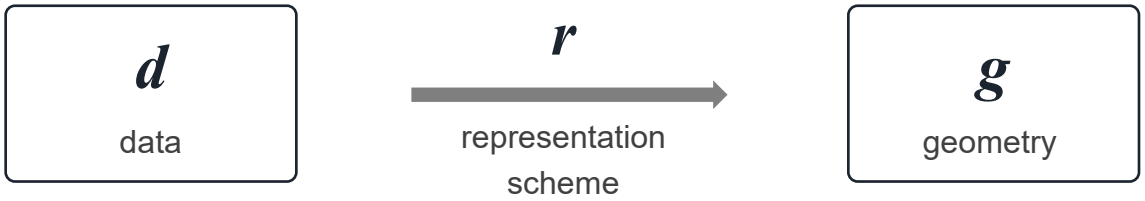
Computational design | Representation



What is a representation?

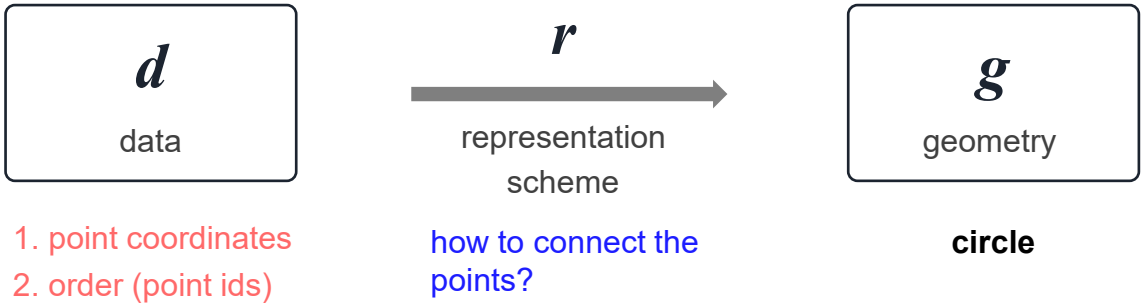
The **digital encoding** of geometric shapes, objects, and spatial relationships using **mathematical data structures** so a computer can store, process, analyze, and render them

Examples of the data structures: Meshes (vertices, edges, faces), Boundary Representations (BReps), Constructive Solid Geometry etc.

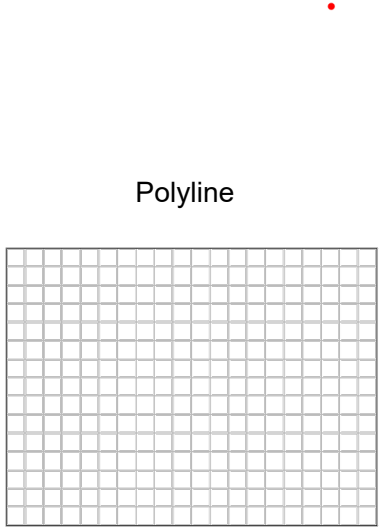


Source: Requicha, A.A.G.: Representations for Rigid Solids: Theory, Methods, and Systems, 1980.

A simple example – Digital representation of a circle



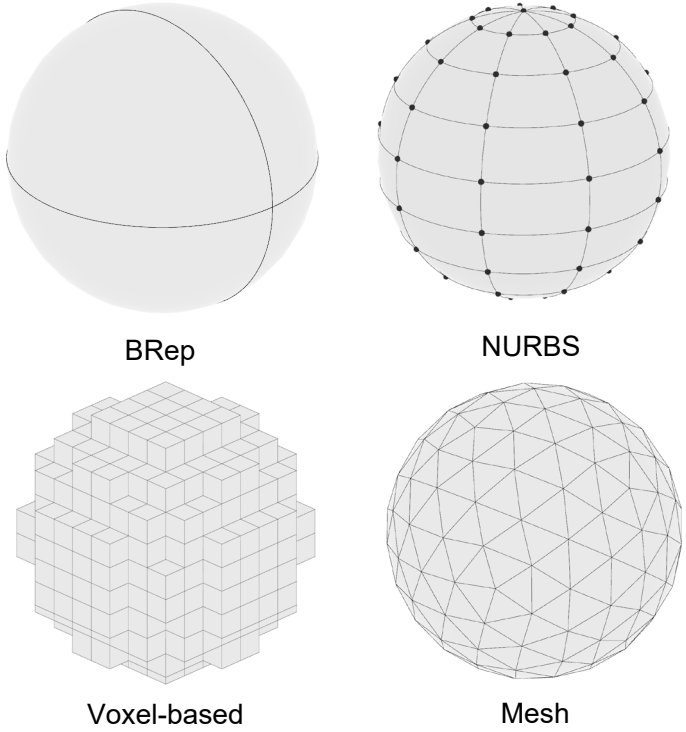
There are multiple ways to represent a geometry…



Polyline

Pixels (Implicit)

Circle



BRep

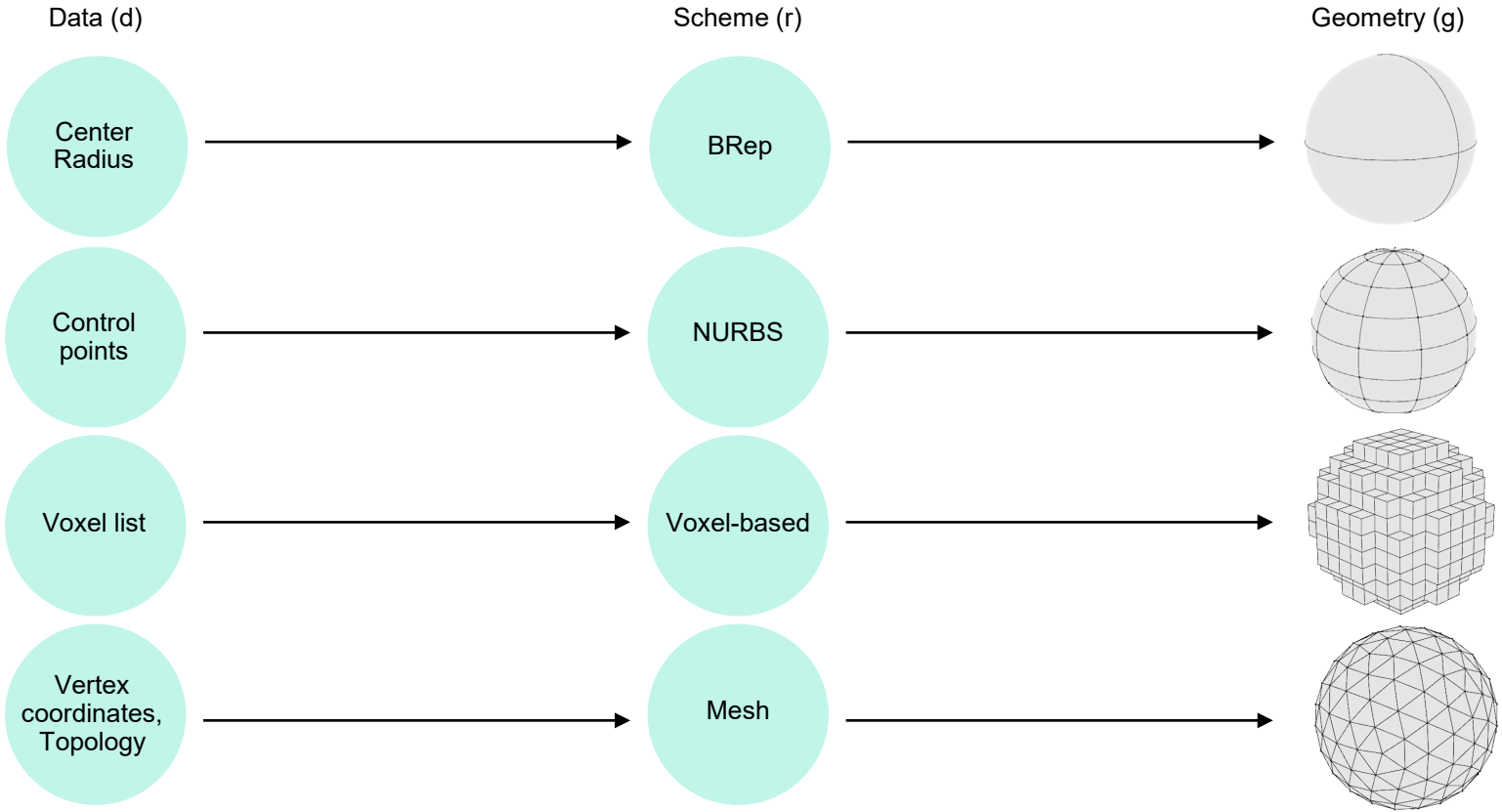
NURBS

Voxel-based

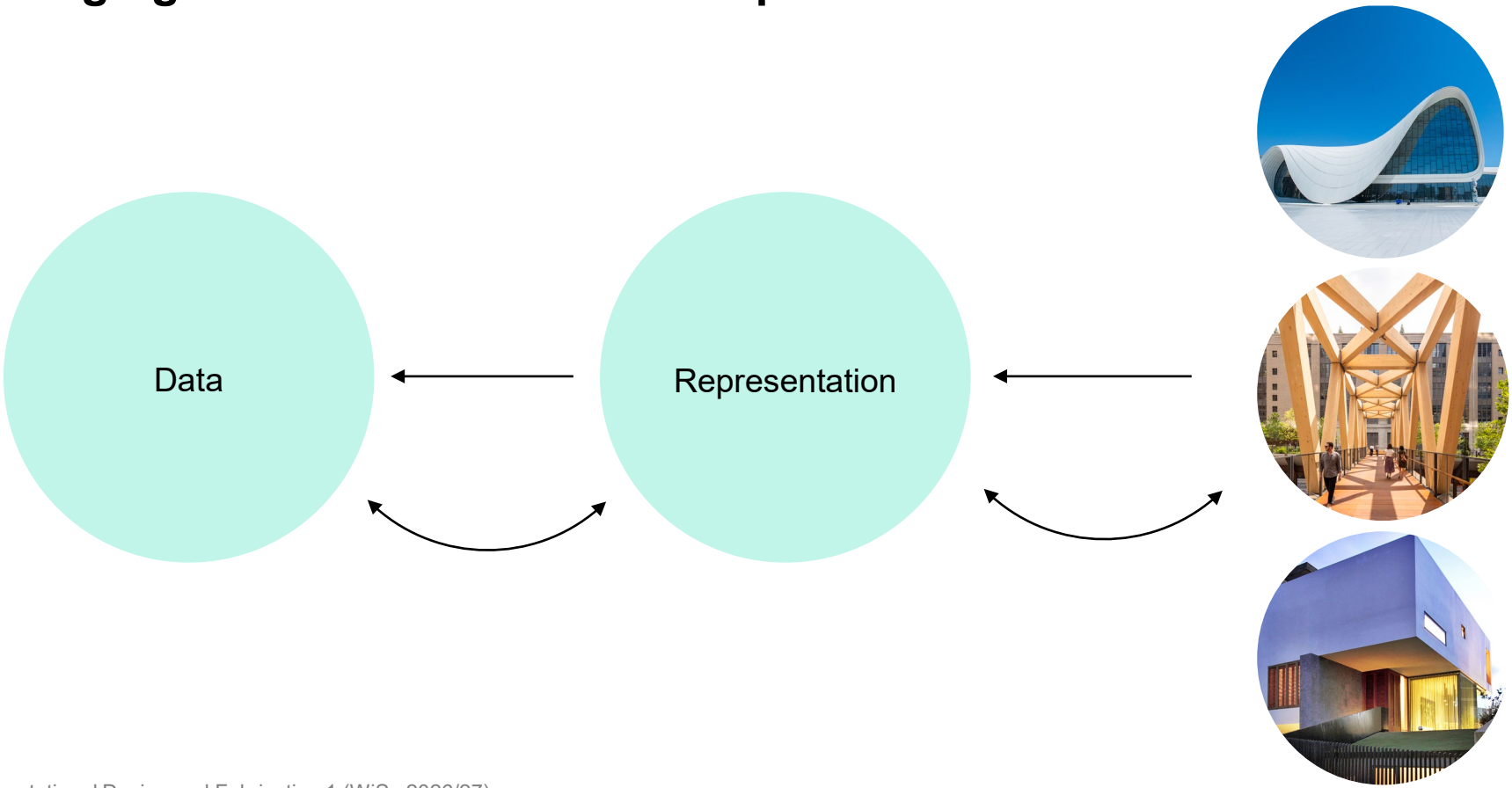
Mesh

Sphere

There are multiple ways to represent a geometry…



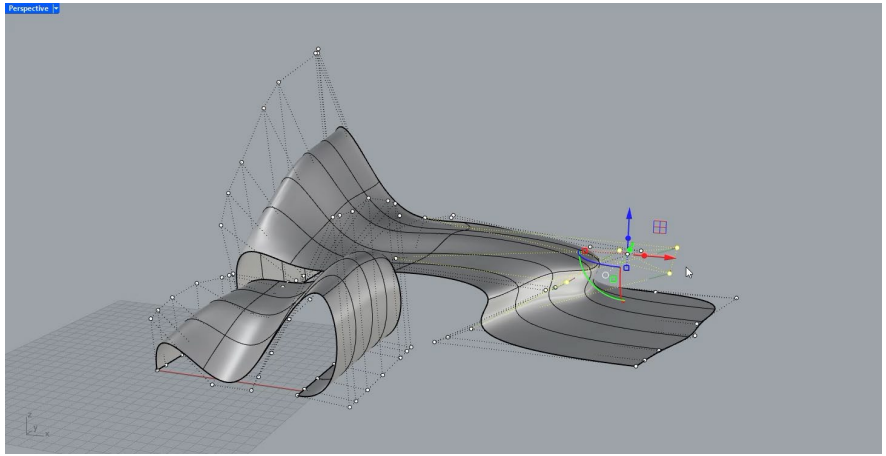
Design goal defines the suitable representation



Level of detail defines the suitable representation

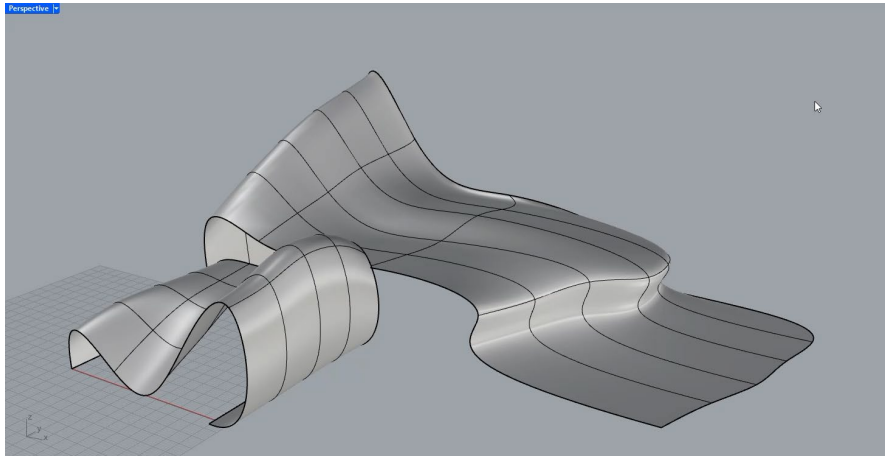


Example: Heydar Aliyev Center by ZHA Architects



Overall shape

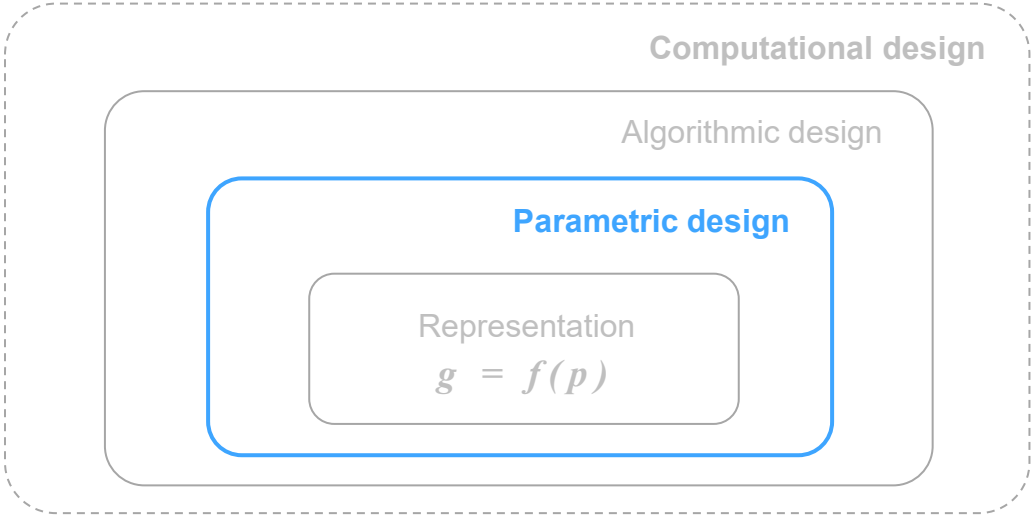
NURBS modelling | Higher level



Panelization

Mesh modelling | Lower level

Computational design | Parametric design



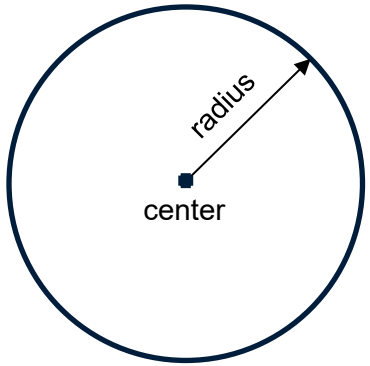
What is parametrization?

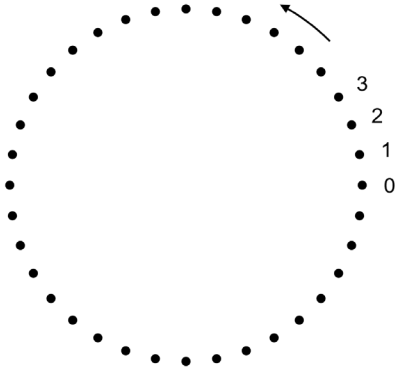
Parametrization refers **to declaring a few independent variables (parameters p),** dependencies and relationships (mapping m) such that all other data (d) can be written as functions of them.

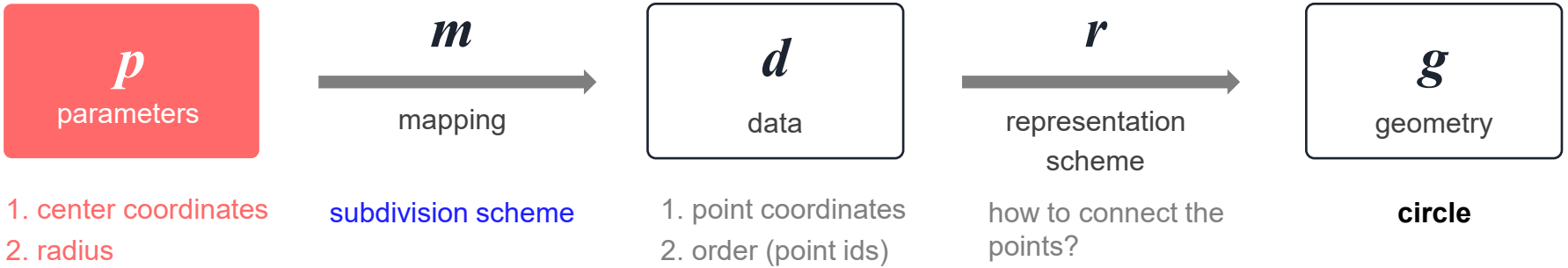
$$d = m(p)$$



Parametrizing a circle…







What is parametric design?

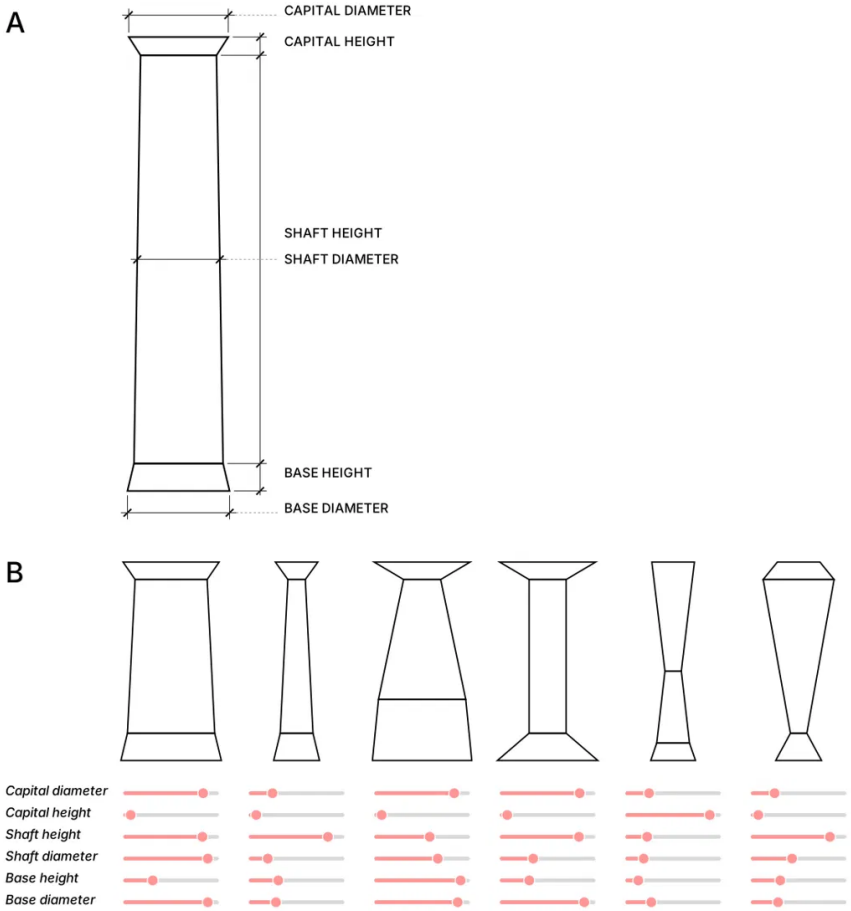
Parametric design is a modelling approach in which design outcomes are defined through **explicit relationships between input parameters and resulting geometric or performance characteristics**.



What is parametric design?

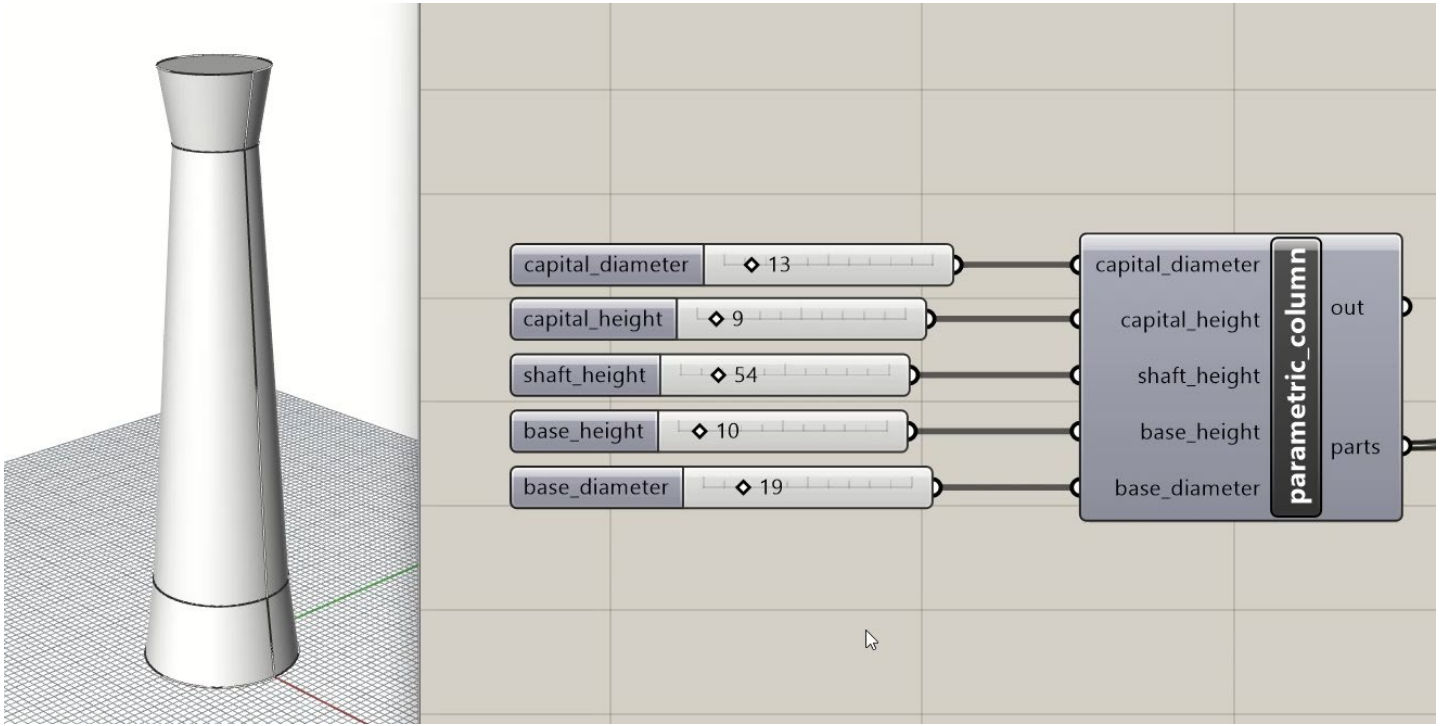
Parametric design is a modelling approach in which design outcomes are defined through **explicit relationships between input parameters and resulting geometric or performance characteristics**.

Example:
Column



Source: Barrios Hernandez, C. R. (2006). Thinking parametric design: Introducing parametric Gaudi. Design Studies, 27(3), 309–324. https://doi.org/10.1016/j.destud.2005.11.006

Parametric model of a column



Parametric design space

PARAMETERS

shaft_height (h_s)

shaft_diameter (d_s)

column_height (h_C) = fixed

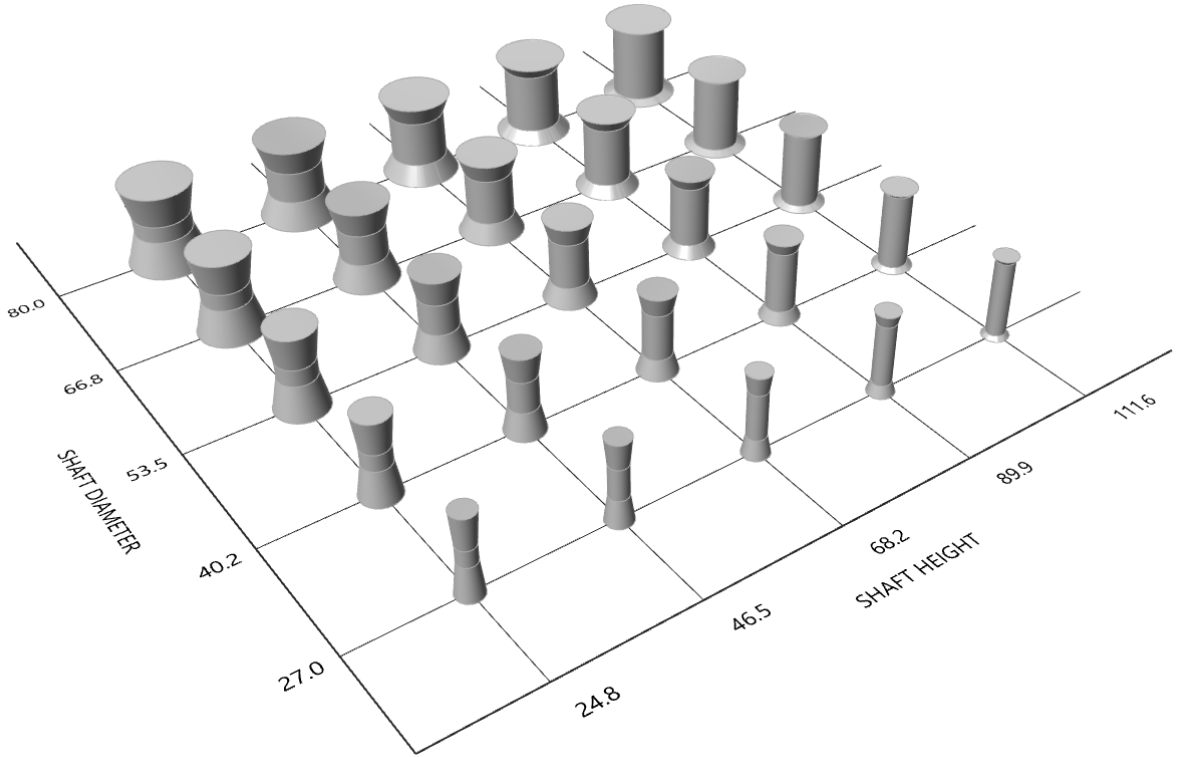
RELATIONSHIPS

base_height (h_b) == capital_height (h_c)

base_diameter (d_b) = 1.5 x shaft_diameter (d_s)

capital_diameter (d_c) = 1.5 x shaft_diameter (d_s)

Design space = all shapes g for every allowed p.



Parametric design space + Performance-based criteria

PARAMETERS

shaft_height (h_s)

shaft_diameter (d_s)

column_height (h_C) = fixed

RELATIONSHIPS

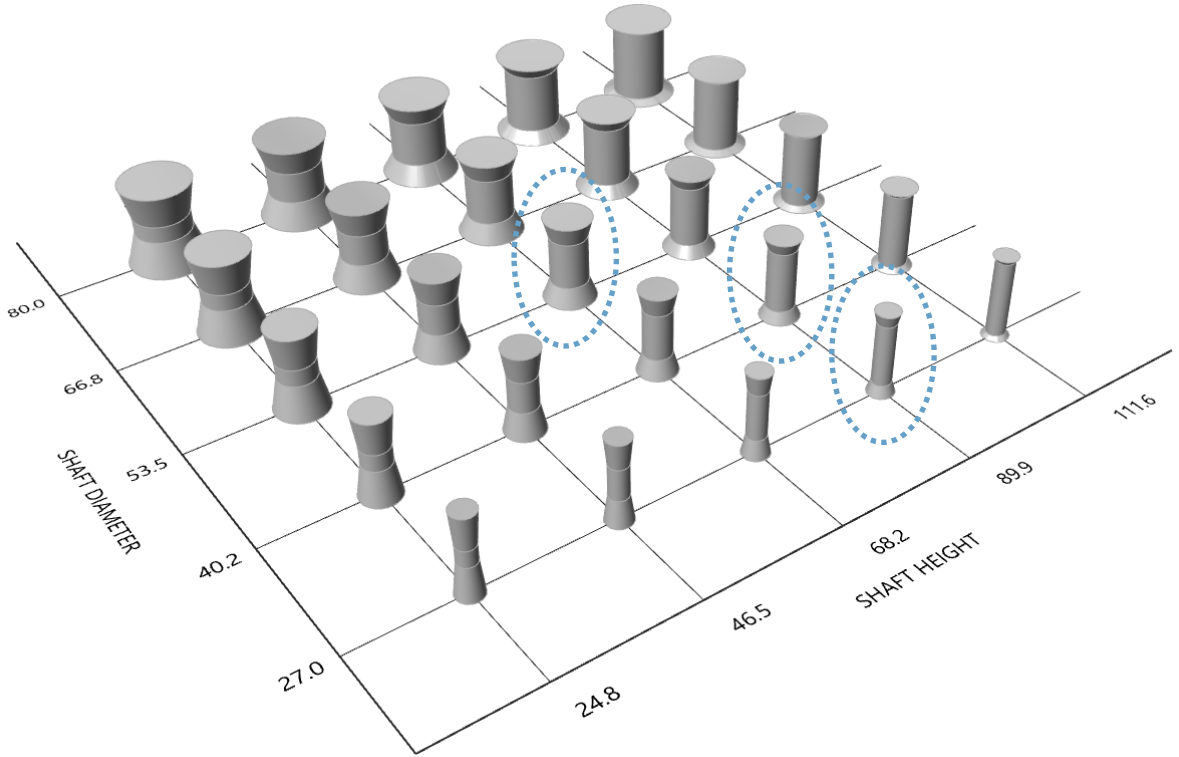
base_height (h_b) == capital_height (h_c)

base_diameter (d_b) = 1.5 x shaft_diameter (d_s)

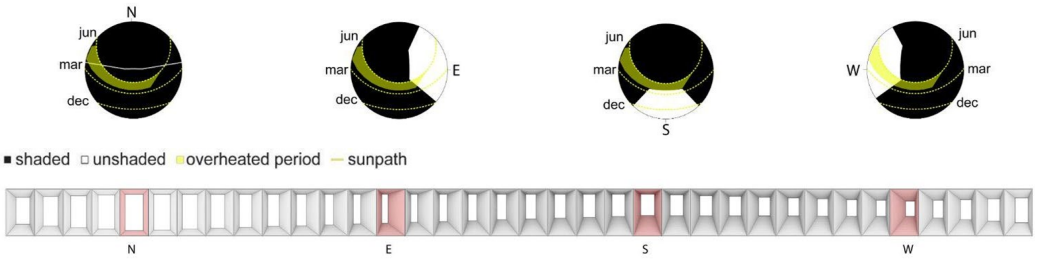
capital_diameter (d_c) = 1.5 x shaft_diameter (d_s)

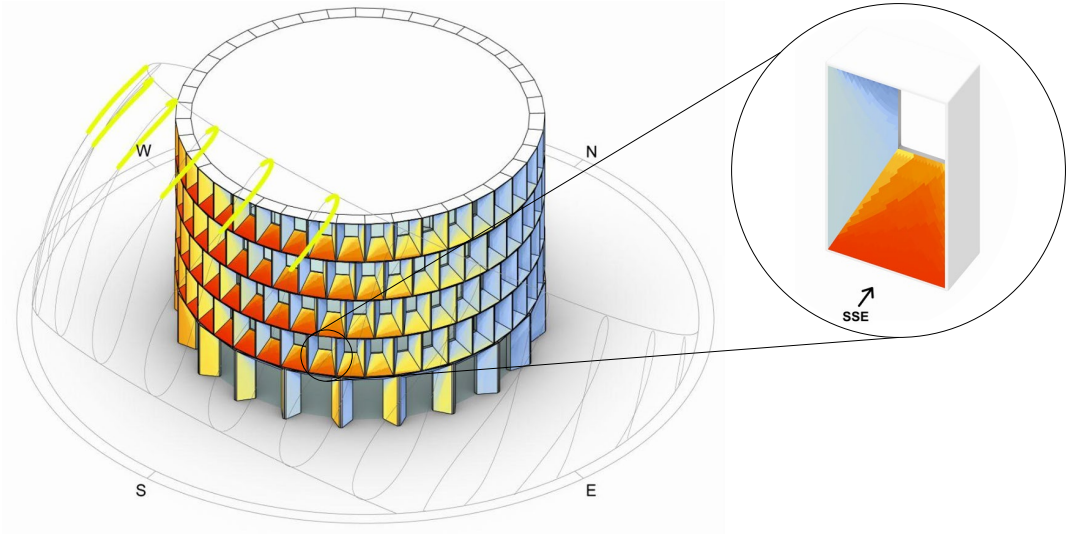
SELECTION CRITERIA

mass, design requirements etc.



Performance-based parametric design

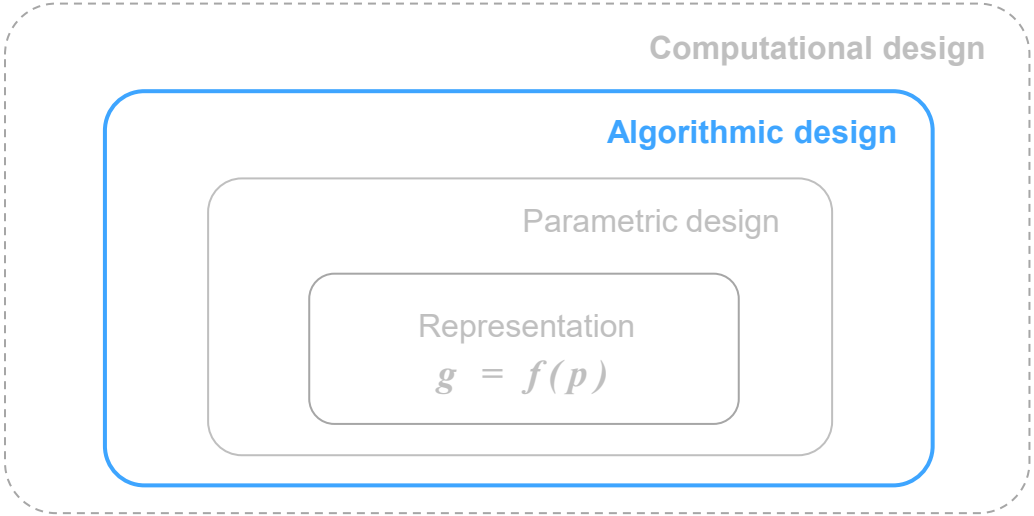






Performance as a **design driver**.

Computational design | Algorithmic design



What is algorithmic design?



What is algorithmic design?



What is algorithmic design?

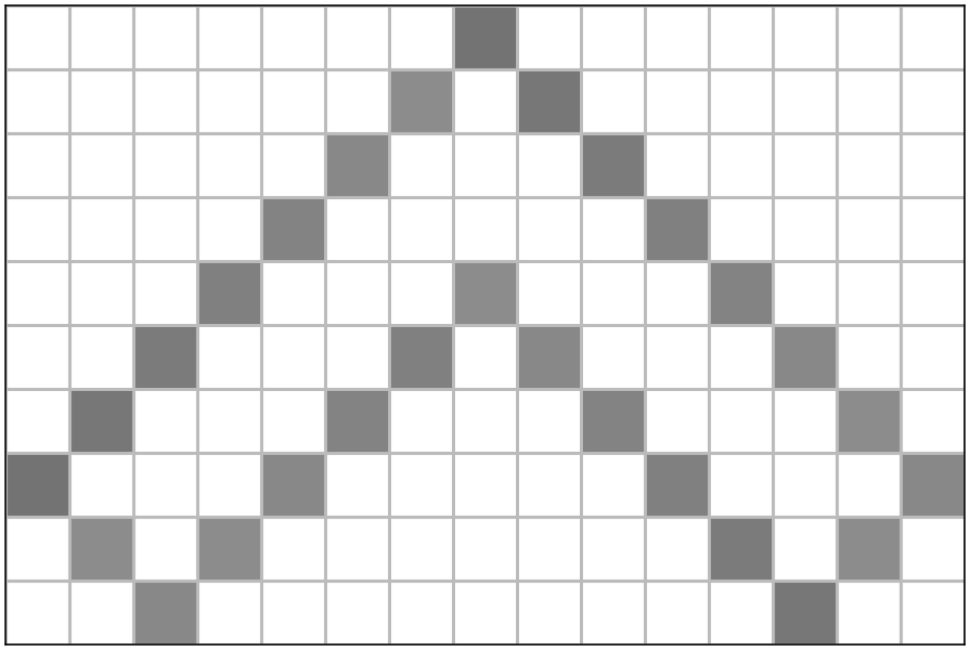
An algorithm is **any well-defined computational procedure** that takes a set of values as input and produces a set of values as output in a finite amount of time.

Algorithmic design refers to the **development of rules and computational procedures,** such as a sequence of algebraic, analytic, and geometric operations, for the manipulation of data and its translation into architectural geometry.



Sources: Cormen, T.H. et al.: Introduction to Algorithms (4th ed.), 2022. Kotnik, T.: Digital Architectural Design as Exploration of Computable Functions, 2010

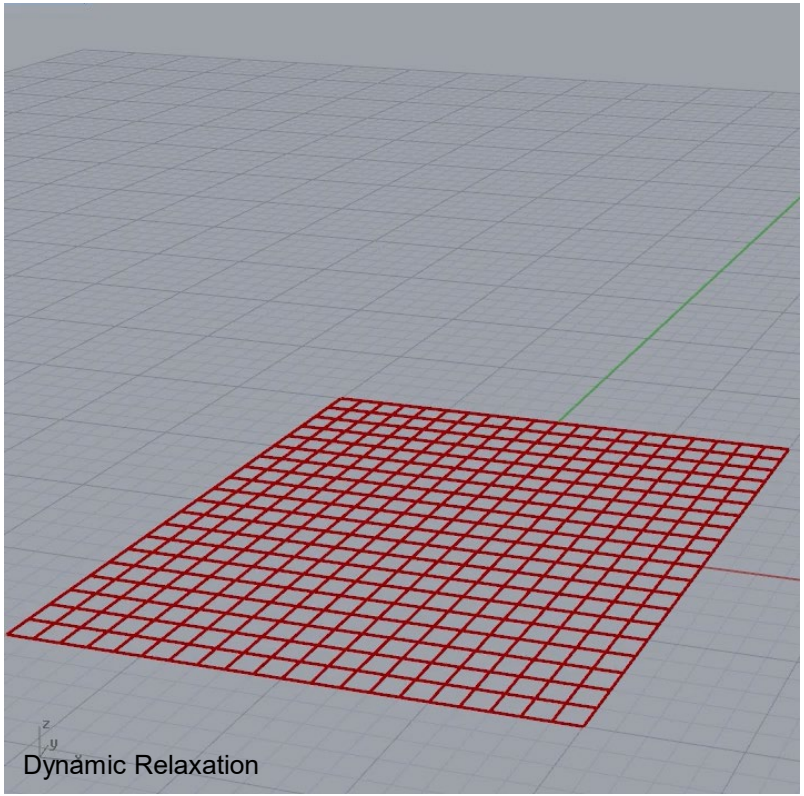
How can you construct this shape algorithmically?



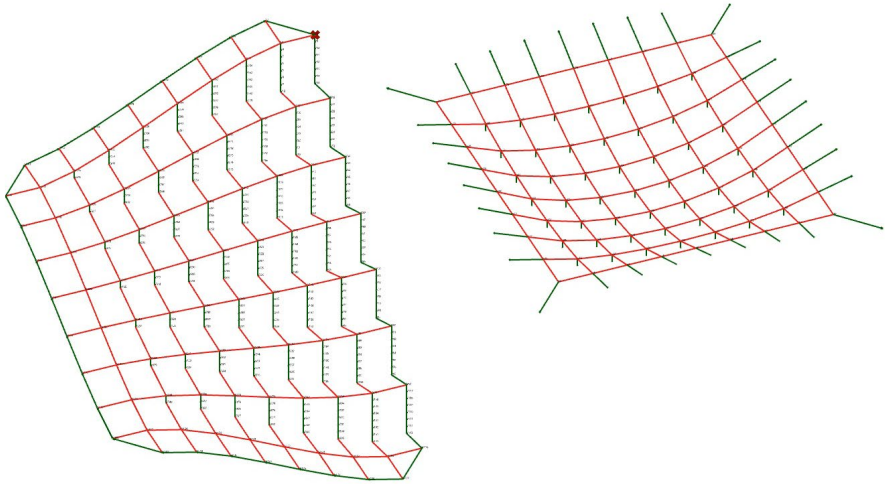
How can you construct this shape algorithmically?



Algorithmic design example – Form-finding



Dynamic Relaxation



Vector-based Graphic Statics

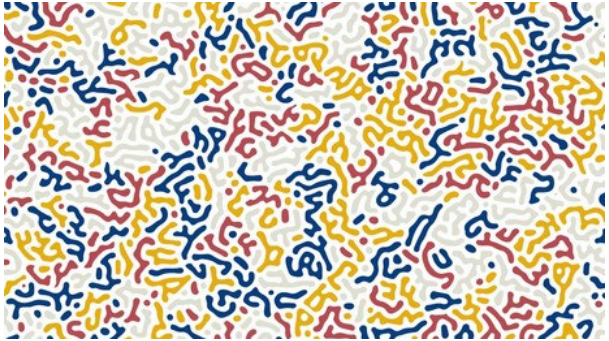
More examples of design algorithms



Geometrically motivated



Physically motivated

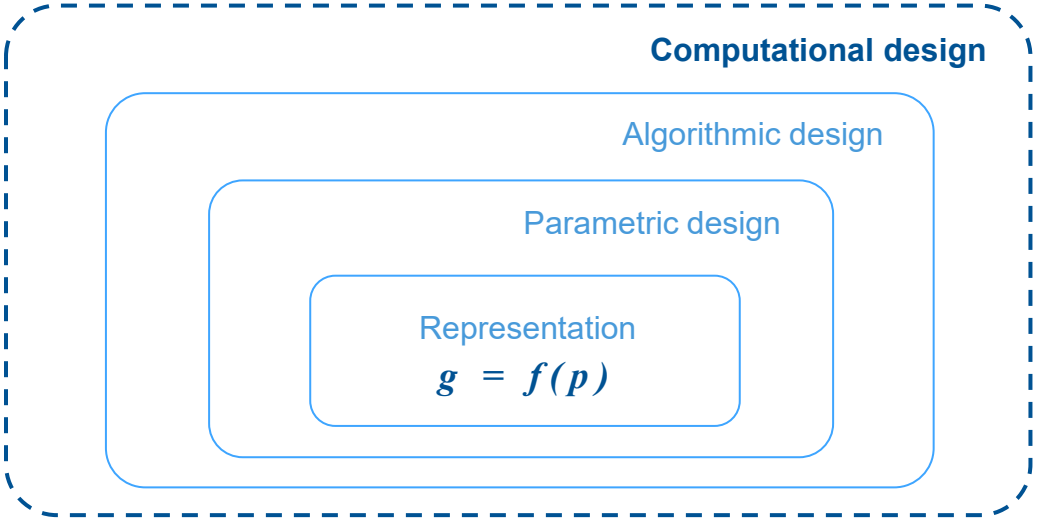


Generative



Optimization

Computational design



Representation — the designer edits **g** directly

Drawing and modelling the shape itself

Parametric — the designer sets **p** of a given f

Parameters, Dependencies and Relations in a fixed model

Algorithmic — the designer writes **f**

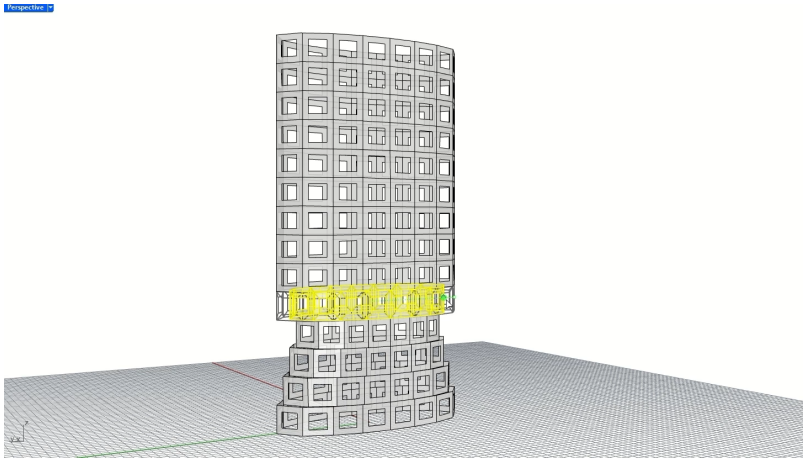
Rules, procedures, simulations

Design approach that uses **computation** to generate, transform, evaluate, and refine design solutions through formalized relationships, quantifiable properties, and algorithmic processes.

Source: Terzidis, K.: Algorithmic Architecture, 2006.

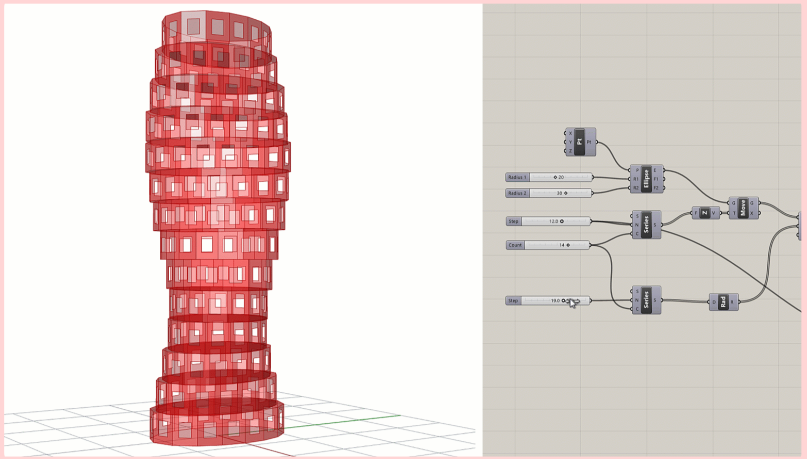
Computational, not computer-aided

Computer-aided

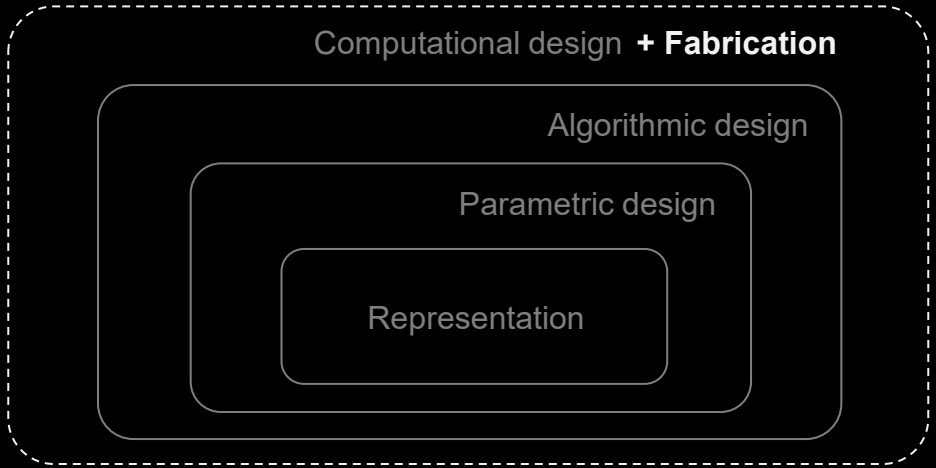


The user defines the design, while the computer translates the user inputs (e.g. clicks and dimensions) through a defined representation into digital geometry and stores the data as a **drafting tool**.

Computational



The computer generates and explores designs from rules and parameters. It participates in generating and evaluating the design, making it a **design tool.**



3 /

Integrated Design and Fabrication Representations

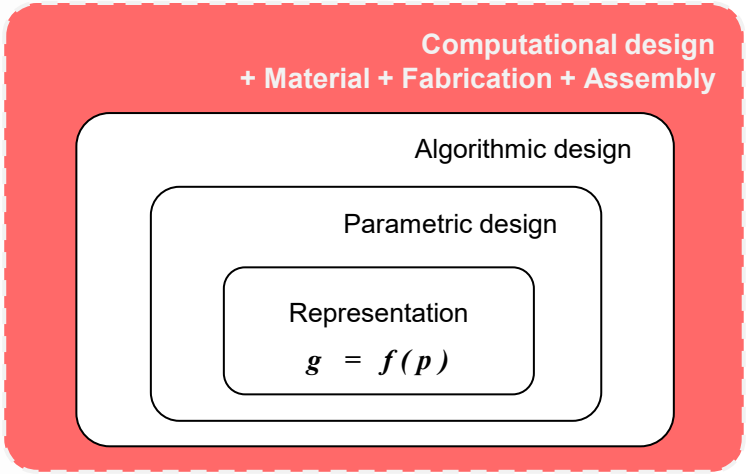
Advanced Integrated Representations for Computable Processes

BUILT TOTAL:
0

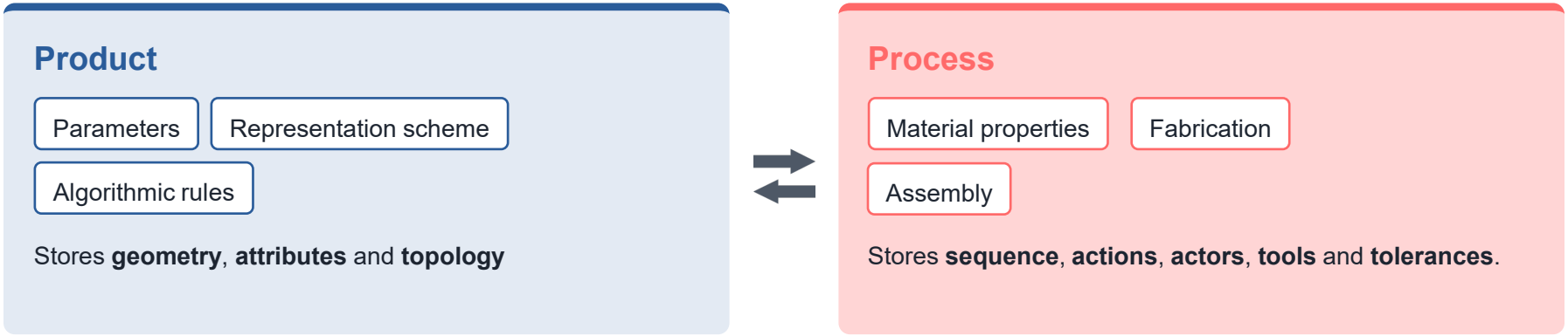
BUILT BY HUMAN:

BUILT BY ROBOT:





What is an integrated representation?



An **integrated representation** is a complete data structure that holds both information of product and process:

Geometric data comprising the parameters and algorithmic rules that define the shape.

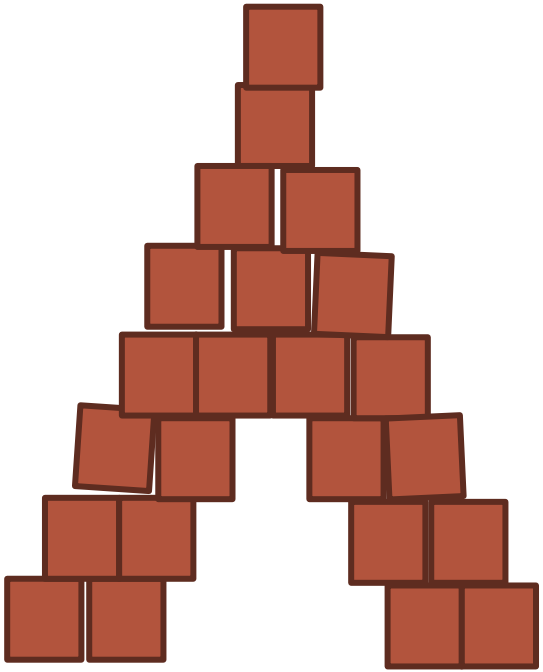
Fabrication data comprising the toolpaths, targets, sequences and material information that define the assembly model.

Sources: Eastman, C.M.: Building Product Models, 1999. Gramazio, F., Kohler, M.: Digital Materiality in Architecture, 2008. Knippers, J., Kropp, C., Menges, A., Sawodny, O., Weiskopf, D.: Integrative computational design and construction: Rethinking architecture digitally, 2021, Fig. 1.

Thinking in materials

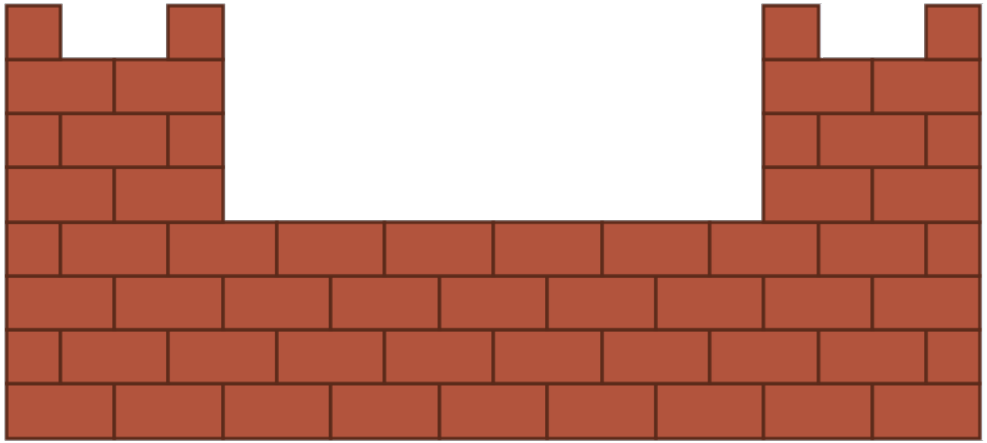


Product

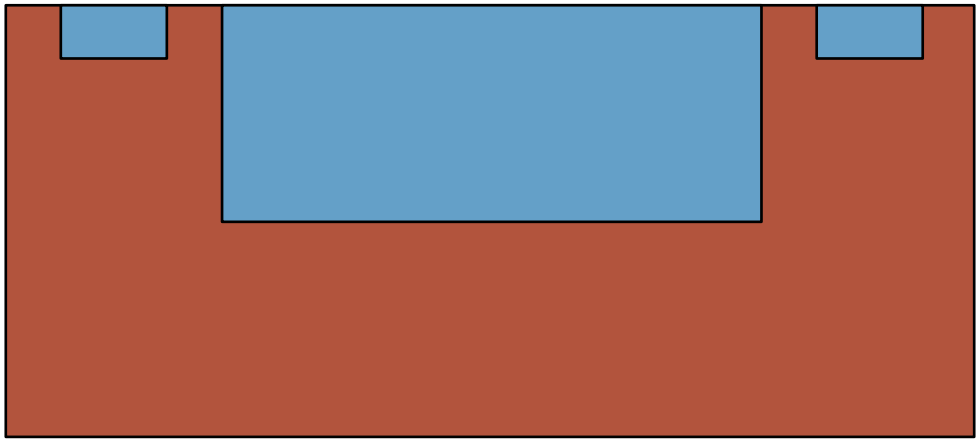


Process

Computational Design of a Castle



Boolean Operators



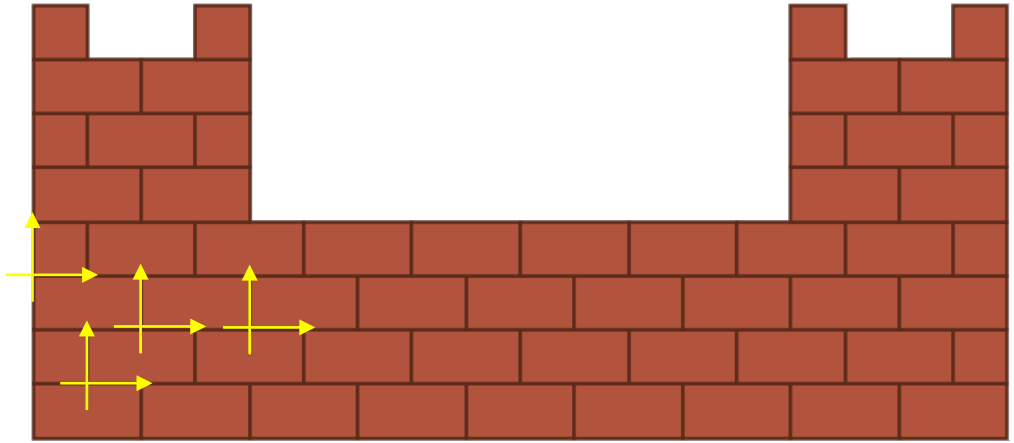
„Brick“ representation



Computational Design of a Castle

Geometry Representation 1

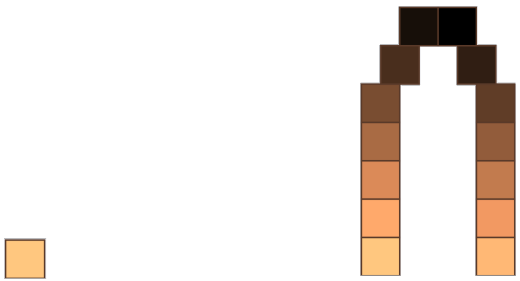
(f,x_1,y_1)
(h,x_2,y_2)
(f,x_3,y_3)
(f,x_4,y_4)

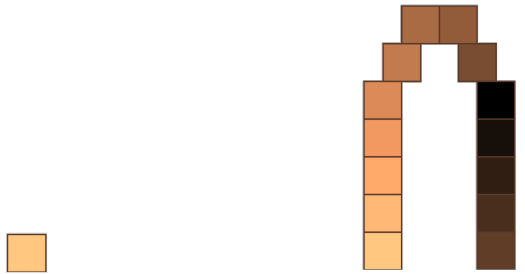


Geometry Representation 2

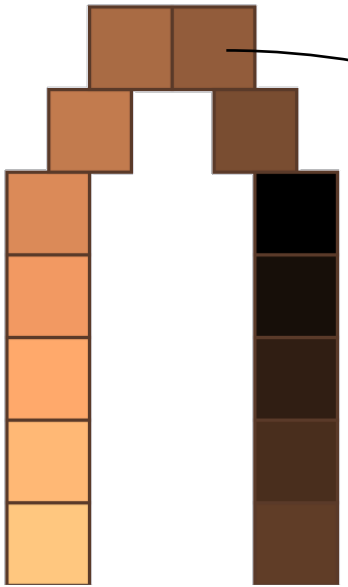
(m,9)
(m,9)
(m,2,v,5,m,2)
…

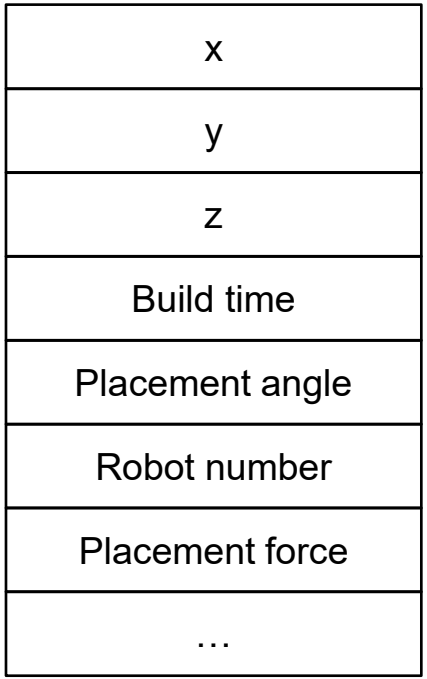
Example: build sequence





There are more dimensions….

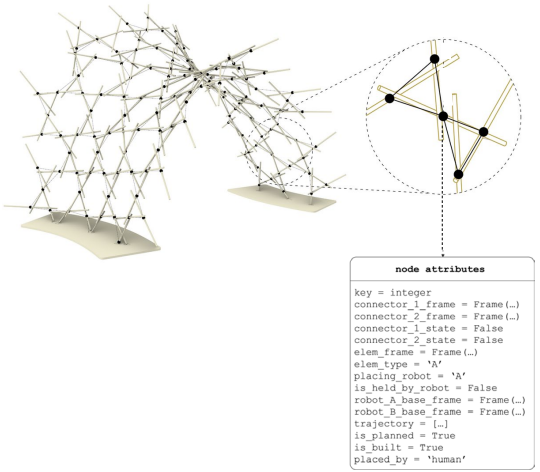





Case study

Computational design

1/ REPRESENTATION SCHEME

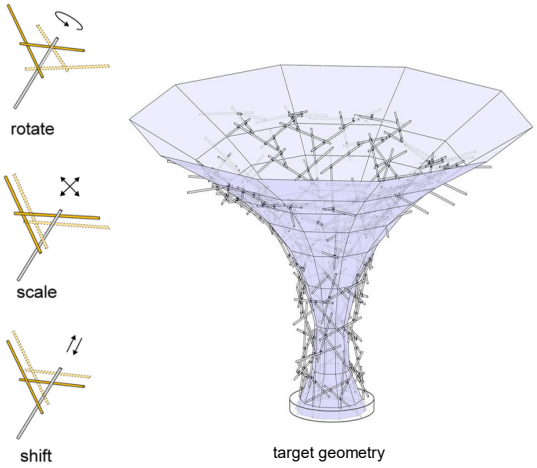


Directed graph

Nodes = rods with attributes

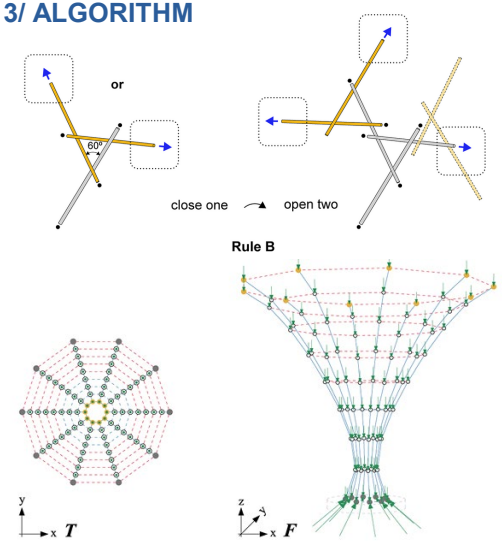
Edges = connectors and task order.

2/ PARAMETERS



Unit parameters. rotation angle, shift, scale, mirror. **Global parameters.** target surface, supports, start rods.

3/ ALGORITHM



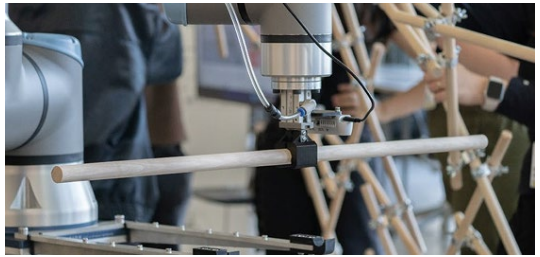
Making grammar. Rules of assembly.

Form-finding. Using CEM to form-find the target geometry

Source: Atanasova, L., Kenny, J.C., Alexi, E.V., Chovghi, F., Mitterberger, D., Dörfler, K.: Building (with) human–robot teams: fabrication-aware design, planning, and coordination of cooperative assembly processes, 2025.

Computational design + Fabrication

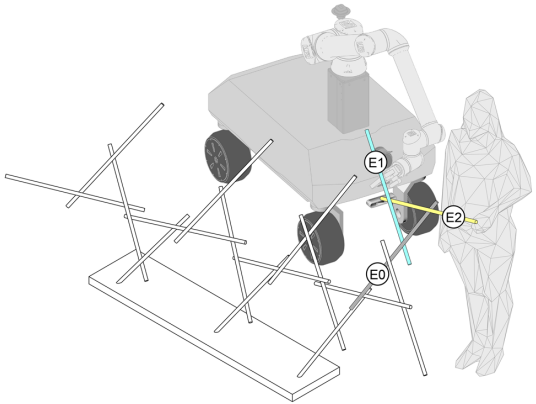
4/ MATERIAL





Timber rods joined by swivel couplers. Three rods lock into one reciprocal-frame unit.

5/ ASSEMBLY PROCESS



Turn-based rod placement, path planning and collaborative assembly.

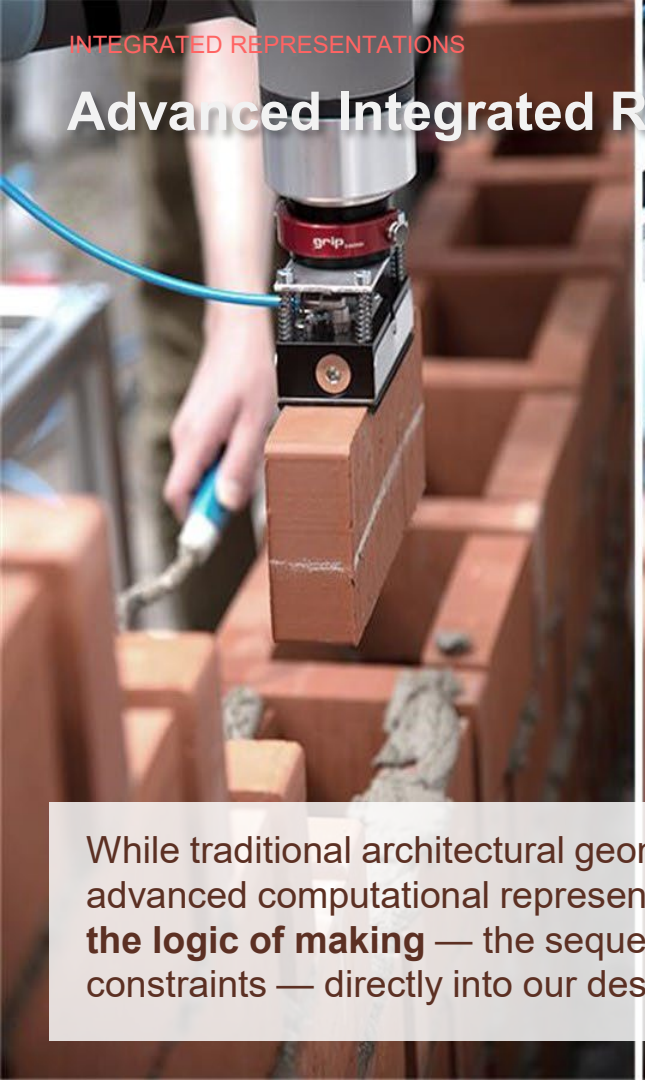
6/ OUTPUT

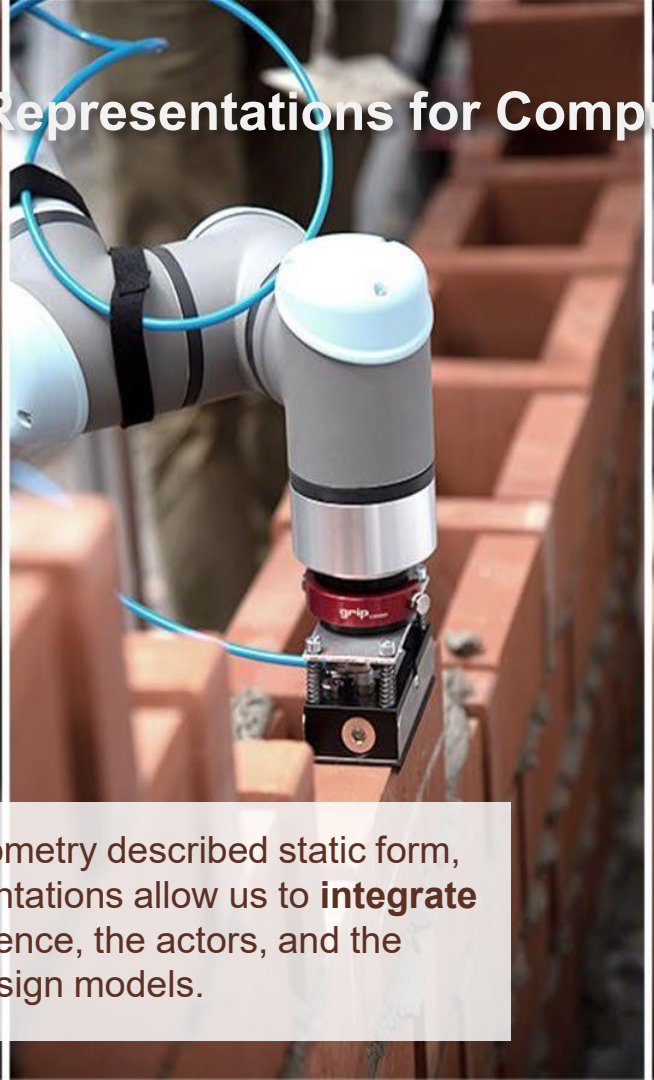


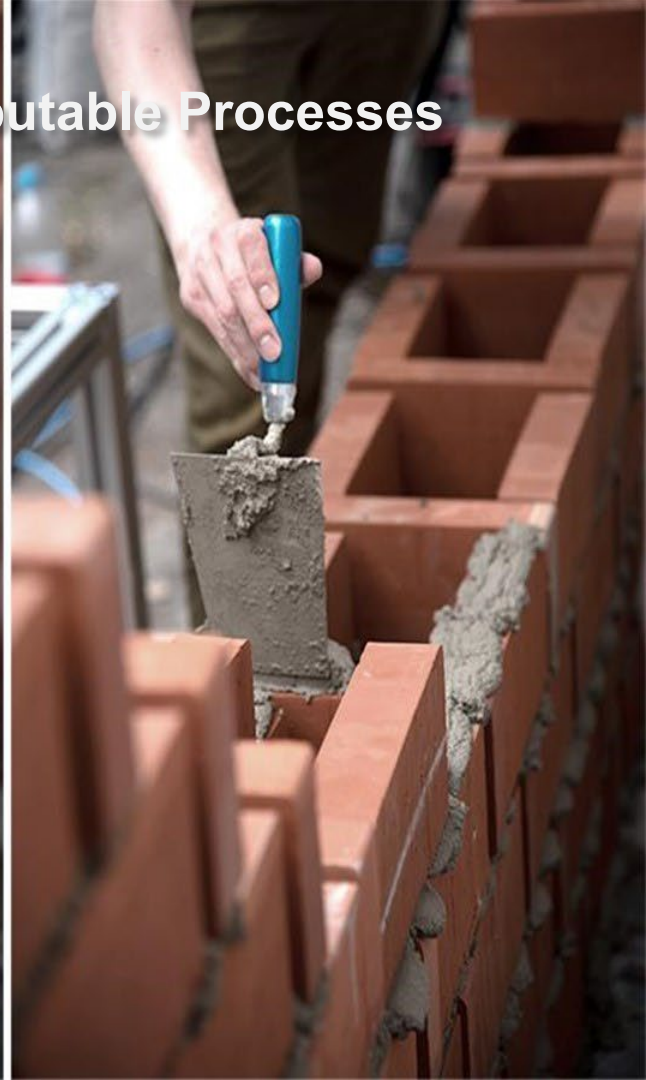
Coordinated **multi-human robot cooperative assembly** using AR interface

Source: Atanasova, L., Kenny, J.C., Alexi, E.V., Chovghi, F., Mitterberger, D., Dörfler, K.: Building (with) human–robot teams: fabrication-aware design, planning, and coordination of cooperative assembly processes, 2025.

Advanced Integrated Representations for Computable Processes







While traditional architectural geometry described static form, advanced computational representations allow us to **integrate the logic of making** — the sequence, the actors, and the constraints — directly into our design models.

4 /

Outlook

PART 1 · PARAMETRIC AND ALGORITHMIC DESIGN



Python basics

up next



Computational geometry

26 Oct



Rule-based design

9 Nov

PART 2 · MATERIALS AND PROCESSES – INTEGRATED REPRESENTATIONS



Sand

23 Nov



Clay

7 Dec



Solid parts

11 Jan



Textile

25 Jan